

Pseudobulk fragments

Group public 10x Genomics PBMC single-cell ATAC fragments into corrected pseudobulk footprint outputs.

① Single-cell ATAC input

Fragments.tsv.gz

chrom	start	end	barcode
chr1	100.6k	100.7k	AAAC...
chr1	105.2k	105.4k	AAAC...
chr2	150.1k	150.3k	AAGT...
chr3	75.9k	76.1k	CTTG...

Metadata

barcode	cell_type
AAAC...-1	T_NK_cell
AAGT...-1	B_cell
CTTG...-1	Monocyte

② Group by cell label

B cell



barcodes

AAAC...
AAGT...
CTTG...

Monocyte



barcodes

AAAC...
AAGT...
CTTG...

T/NK



barcodes

AAAC...
AAGT...
CTTG...

Each barcode contributes fragments to one retained group.

③ Aggregate and QC



fragments → pseudo-BAM

B cell



fragments → pseudo-BAM

Monocyte



fragments → pseudo-BAM

T/NK

QC filters

min cells 300
min fragments 50k
filter low-depth

Options

tabix index
chrom filters
barcode cleanup

④ Corrected outputs

Manifest / QC summary

group	status	cells	outputs
B_cell	pass	1,102	bam/bw/tsv
Monocyte	pass	1,045	bam/bw/tsv
other	filtered	112	-

Primary fp-tools outputs

corrected cut-site bw

footprint-score bigWig

diff. report

volcano + KNN UMAP

○ single cell barcode — pseudobulk fragment  corrected footprint tracks